

SHYAM SARAN P

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Portfolio



github.com/shyamsaran348



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SUMMARY

Final-year IT student experienced in building full-stack web applications and AI-integrated systems. Skilled in backend development, RAG-based document search, and machine learning with Python.

EDUCATION

SSN College of Engineering, Chennai

Expected May 2027

B.Tech in Information Technology

CGPA: 7.9

Doveton Boys Higher Secondary School, Chennai

2023

Higher Secondary (Class XII)

90.2%

EXPERIENCE

Junior AI Analyst (Part-Time) A2000 Solutions

Jun 2026 – Present

- Developed an educational mentor AI assistant by fine-tuning a **Small Language Model (SLM)** using **QLoRA**.
- Created synthetic **JSONL** training datasets and prompt templates to improve the assistant's response quality.

Software Development Intern Highways Department, Government of Tamil Nadu

May 2026 – Jun 2026

- Developed a full-stack portal using **React**, **FastAPI**, and **PostgreSQL** to streamline highway tender and administrative workflows.
- Implemented secure **JWT** authentication, hierarchical **Role-Based Access Control (RBAC)**, and detailed audit logs.

Web Developer Intern A2000 Solutions

Jun 2025 – Jul 2025

- Built a complete Learning Management System (LMS) using **Flask** and **PostgreSQL** with role-based access for students, instructors, and administrators.
- Developed **REST APIs** for course management and learning workflows, optimizing SQL queries to improve application performance.

PROJECTS

VeriSource AI *Python, FastAPI, PostgreSQL, ChromaDB, ONNX, React*

GitHub

- Built a **RAG**-based platform for educational institutions to search rules and regulations documents and find relevant information.
- Implemented document search using **ChromaDB** and **ONNX Runtime**, reducing memory usage by **60%**.

SDGSync – Enterprise Project Alignment Platform *Python, FastAPI, PostgreSQL, React*

GitHub

- Built a web platform to categorize project proposals based on **UN Sustainable Development Goals (SDGs)**.
- Used a **BERT**-based classifier with a RAG pipeline, achieving **95%** classification accuracy.

AI-Powered Examination Portal (Ongoing) *FastAPI, React.js, PostgreSQL*

GitHub

- Developed an examination portal with automated **MCQ generation** and assessment using **FastAPI**, **React**, and **PostgreSQL**.
- Currently building a **PyTorch**-based proctoring engine to analyze **facial expressions**, **gaze direction**, **head pose**, and **audio signals** for real-time behavioral monitoring.

SKILLS

Languages

Python, Java (Intermediate), C

Frameworks & APIs

FastAPI, Flask, React (Basic), RESTful APIs

AI Integration

Retrieval-Augmented Generation (RAG)

Databases & Tools

PostgreSQL, MySQL, SQLite, Git, GitHub, VS Code

Concepts

Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PUBLICATIONS

Reddit Rule-Violation Classification Using LLM-Assisted Data Augmentation

IEEE ICDSAAI 2026

Third Author DOI

Explainable and Robust Heritage Intelligence Framework (ERHIF)

International Journal of Advanced Science and Engineering (IJASE), 2026

Fourth Author DOI